Conjecture 00000003161: two exact measurements for all one-sparse signals

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1 The missing recovery assumption

The conjecture states an underdetermined linear system $y = Ax$ and claims a uniform necessary measurement count $m \geq Ck \log(N/k)$. The constant in such a uniform lower bound must be positive and independent of N. No stability, measurement noise, finite precision, restricted decoder or computational constraint is imposed. For exact recovery over the real numbers this necessary bound is false. We prove that for every $C > 0$ some $N > 2$ admits uniform recovery of all one-sparse real signals from two measurements, while $2 < C \log N$.

The distinction between exact real arithmetic and additional precision requirements is standard. See Ardestanizadeh–Cheraghchi– Shokrollahi, *Bit Precision Analysis for Compressed Sensing* (2009), introduction: Vandermonde measurements require only $2k$ exact measurements, while finite precision adds a separate resource question. Our proof below is self-contained for $k = 1$. It does not refute noise-robust or stable recovery bounds with their additional assumptions. The undefined noise-sphere packing clause does not supply a noise model or a stability requirement for the displayed necessary count.

2 A genuine sensing matrix and all signals

For every $N \geq 3$, index coordinates by $j = 0, \ldots, N-1$ and define

$$A_N = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 0 & 1 & \cdots & N-1 \end{pmatrix}.$$

It is a real $2 \times N$ matrix, with fewer measurements than coordinates. Every vector with at most one nonzero coordinate is $x = te_i$ for some index i and real t, including the zero vector when $t = 0$. Its actual matrix product is

$$A_N(te_i) = (t, it).$$

If $A_N(te_i) = A_N(ue_j)$, the first coordinate gives $t = u$. If $t = 0$, the signals both vanish. Otherwise the second coordinate gives $it = jt$, hence $i = j$. Thus the measurement map is injective on the entire set of one-sparse signals, including zero, uniformly in the index and coefficient.

3 A total decoder and the actual minimum

For any measurement $y \in \mathbf{R}^2$ which has a one-sparse preimage, let $D_N(y)$ be that preimage; it is unique by the preceding proof. For all other y, define $D_N(y) = 0$. This is a total map and

$$D_N(A_N x) = x \qquad \text{for every one-sparse } x.$$

No computability or continuity requirement was present in the problem. The formal decoder uses classical choice only to select the unique preimage on this range and proves correctness from injectivity.

Define $\mathrm{CanRecover}_N(m, k)$ to mean that there exist an actual real $m \times N$ matrix and a total decoder recovering every vector with support cardinality at most k. The construction proves $\mathrm{CanRecover}_N(2, 1)$. Hence the genuine least measurement count

$$m_{\min}(N, 1) = \min\{m \in \mathbf{N} : \mathrm{CanRecover}_N(m, 1)\}$$

exists and is at most two. We do not assume or assign the value of this minimum and do not need to prove equality with two.

4 Every positive uniform constant fails

Fix any $C > 0$. Choose a natural number n with $n > \exp(2/C)$ and put $N = n + 3$. Then $N > 2$ and

$$m_{\min}(N, 1) \leq 2 < C \log N = C \cdot 1 \cdot \log(N/1).$$

The actual minimum therefore violates the proposed necessary bound. The argument works for every positive dimension-independent C; choosing a smaller constant cannot repair the uniform assertion. Allowing C to depend arbitrarily on N, or adding a specified noise/stability model, would give a different proposition.

5 Lean correspondence and verification

Signals are all functions `Fin (n+3)` to $\mathbf{R}$. Sparsity is the cardinality of the actual filtered support. `sparse_one_single` proves the complete single-coordinate representation from that condition. The measurement is Mathlib's `Matrix.mulVec` for the displayed matrix, and its action is proved from matrix multiplication. `decoder_correct` applies to all one-sparse signals. `minimalMeasurements` uses `Nat.find` on the proved nonempty class of successful matrices and decoders. The final `exact_recovery_counterexample` quantifies over every $C > 0$, exhibits $N = n + 3 > 2$, and gives both an actual successful two-row matrix and the violated minimum bound.

Lean 4.33.0 and Mathlib commit `db584cd6d46c92f209a44c0f1c829460d327499d` are pinned. Receipts record a new Main build, transitive axiom audits, and full official `leanchecker --fresh Main` replay. No `sorry`, `native_decide` or additional axiom is used. OpenAI Codex assisted; a separate supervising Codex agent reviewed the formal chain. This is AI review, not human certification. Organizer acceptance is pending.